

Chapter 1

Homomorphisms and Isomorphisms

Consider the two groups $\mathbb{Z}_6$ and $\mathbb{Z}_2 \times \mathbb{Z}_3$ (under addition). $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ and

$$\mathbb{Z}_2 \times \mathbb{Z}_3 = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2)\}.$$

These groups have the same number of elements, but as it turns out they have much more in common. Pair the elements as follows:

$$\begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ (0,0) & (1,1) & (0,2) & (1,0) & (0,1) & (1,2) \end{array}$$

Table 1.1: The elements from $\mathbb{Z}_6$ are mapped to $\mathbb{Z}_2 \times \mathbb{Z}_3$.

This map identifies the zero and identities of each group, $0 \rightarrow (0, 0)$ and $1 \rightarrow (1, 1)$ respectively. Note that in $\mathbb{Z}_6$, $3 + 4 = 1 \pmod{6}$ and in $\mathbb{Z}_2 \times \mathbb{Z}_3$, $(1, 0) + (0, 1) = (1, 1)$. Our map identifies $1 \rightarrow (1, 1)$. Our map can be formulated with $(3x + 4y) \pmod{6} \rightarrow (x, y)$ is a function from $f: \mathbb{Z}_6 \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_3$.

Though these groups appear quite different, this function shows that their behavior is identical. We will call this function an *isomorphism* and two such groups *isomorphic*.

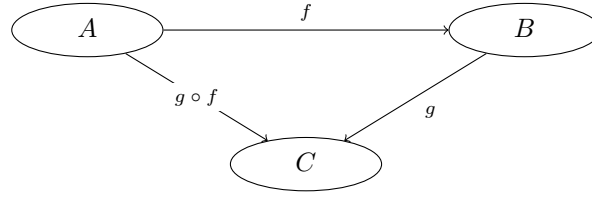
1.1 Functions

We covered functions in MTH 335, but that was a while ago.

Definition 1. A *function* (or *mapping*) from sets A to B , $f: A \rightarrow B$, is a rule that assigns each element in A to exactly one element in B .

- A is called the *domain* of f .
 - B is called the *range* of f .¹
 - If $f(a) = b$ for some $a \in A$, then b is called the *image of a under f*.
 - $\{b: b \in B \text{ and } f(a) = b \text{ for some } a \in A\}$ is called the *image of A under f*.
- Functions are well defined. Meaning that an input is assigned to one and only output.

Definition 2. Let $f: A \rightarrow B$ and $g: B \rightarrow C$. The *composition* of $g \circ f$ is the mapping from A to C defined as follows: $g \circ f(a) = g(f(a))$ for any $a \in A$.


 Figure 1.1: Composition of functions f and g .

Definition 3. A function, $f: A \rightarrow B$, is called *one-to-one*² if $f(a_1) = f(a_2)$ implies that $a_1 = a_2$.

A function $f: A \rightarrow B$ is *onto* if for every $b \in B$ there is an $a \in A$ such that $f(a) = b$.

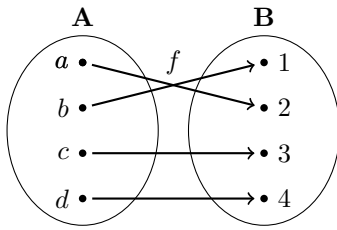


Figure 1.2:

$f: A \rightarrow B$ is one-to-one and $f: A \rightarrow B$ is not one-to-one onto.

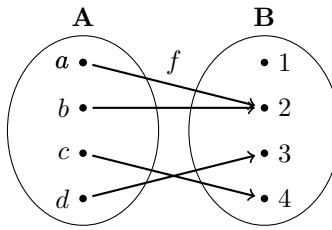


Figure 1.3:

$f: A \rightarrow B$ is not one-to-one and is not onto.

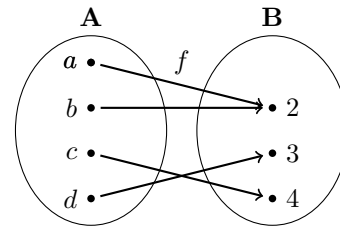


Figure 1.4:

$f: A \rightarrow B$ is onto but not one-to-one.

1.2 Homomorphism

Definition 4. A (ring) *homomorphism*, f , from a ring R to a ring S is a function $f: R \rightarrow S$ that preserves the two ring operations:

$$f(a + b) = f(a) + f(b) \text{ and } f(ab) = f(a)f(b)$$

for $a, b \in R$.

Theorem 1. Properties of Ring Homomorphisms. Let f be a ring homomorphism from rings R to S , and let A be a subring of R .

1. $f(0_R) = 0_S$.
2. $f(-a) = -f(a)$ for every $a \in R$.
3. $f(a - b) = f(a) - f(b)$ for every $a, b \in R$.
4. For any $a \in R$ and any positive integer n , $f(nr) = nf(r)$ and $f(r^n) = f(r)^n$.
5. If R is commutative then $f(R)$ is commutative.
6. $f(A) = \{f(a) : a \in A\}$ is a subring of S .
7. If R has unity 1_R , $S \neq \{0\}$, and f is onto
 - (a) then $f(1_R)$ is the unity of S .
 - (b) if $aa^{-1} = 1_R$ for any $a, a^{-1} \in R$, then $f(a)f(a^{-1}) = 1_S$ and $f(a)^{-1} = f(a^{-1})$.

¹Also called the *co-domain*.

²Also called an *injective function*.

1.3 Isomorphisms

Definition 5. A *ring isomorphism* from a ring R to a ring S is a function that is one-to-one and onto.

Definition 6. If there exists a ring isomorphism $f: R \rightarrow S$ then R and S are said to be *isomorphic*.

Showing that two rings are isomorphic usually requires finding an isomorphism between the, and this can be tricky at times. However looking at the properties from theorem 1 can serve as a guide.